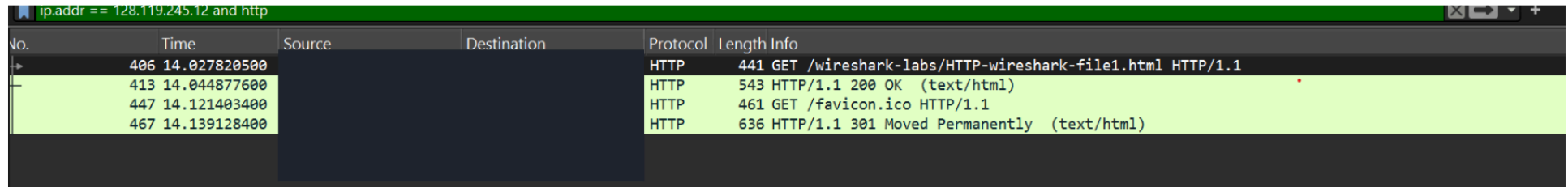


HTTP Wireshark Lab | Question 1

Is your browser running HTTP version 1.0 or 1.1? What version of HTTP is the server running?

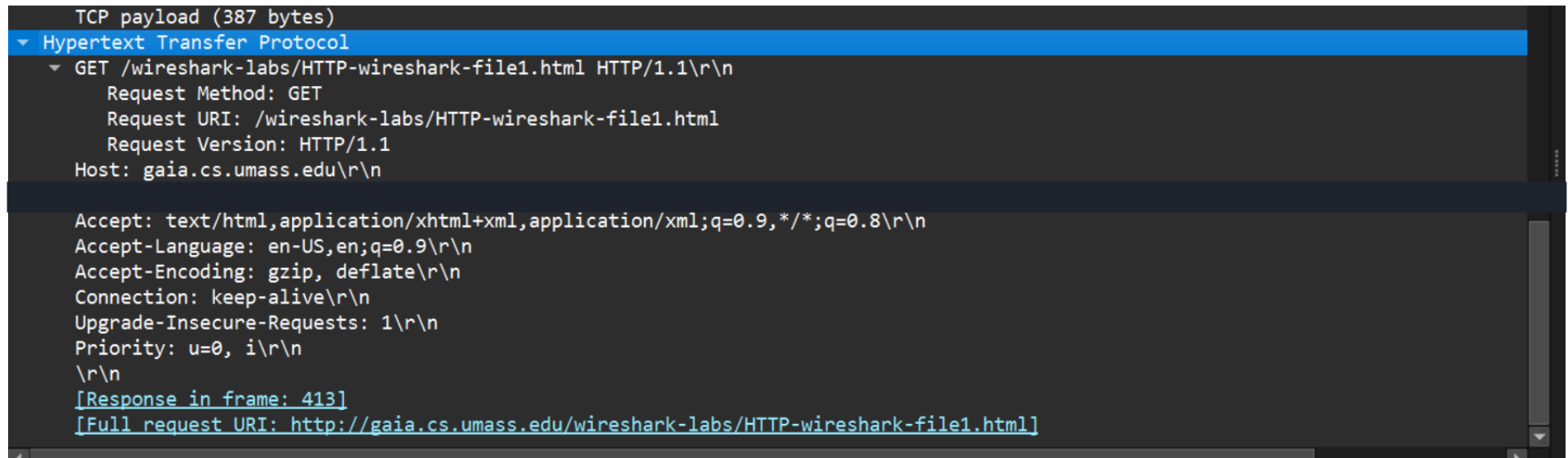
My browser and the server both use HTTP/1.1. The GET request and server response show HTTP/1.1.

The packet list shows HTTP/1.1 in the server response; the request details show HTTP/1.1 in the GET.



A screenshot of the Wireshark packet list. The filter bar at the top shows 'ip.addr == 128.119.245.12 and http'. The packet list table has columns: No., Time, Source, Destination, Protocol, Length, and Info. Four packets are listed, with the first three highlighted in light green. The first packet is a GET request for a file named 'HTTP-wireshark-file1.html'. The second packet is a 200 OK response. The third packet is a GET request for a file named 'favicon.ico'. The fourth packet is a 301 Moved Permanently response.

No.	Time	Source	Destination	Protocol	Length	Info
406	14.027820500			HTTP	441	GET /wireshark-labs/HTTP-wireshark-file1.html HTTP/1.1
413	14.044877600			HTTP	543	HTTP/1.1 200 OK (text/html)
447	14.121403400			HTTP	461	GET /favicon.ico HTTP/1.1
467	14.139128400			HTTP	636	HTTP/1.1 301 Moved Permanently (text/html)



A screenshot of the Wireshark packet details pane for the first packet (No. 406). The pane shows the TCP payload (387 bytes) and the Hypertext Transfer Protocol section. The GET request details are expanded, showing the Request Method (GET), Request URI (/wireshark-labs/HTTP-wireshark-file1.html), Request Version (HTTP/1.1), and Host (gaia.cs.umass.edu). Below the request details, the server's response headers are visible, including Accept, Accept-Language, Accept-Encoding, Connection, Upgrade-Insecure-Requests, and Priority. At the bottom, there are links to the response in frame 413 and the full request URI.

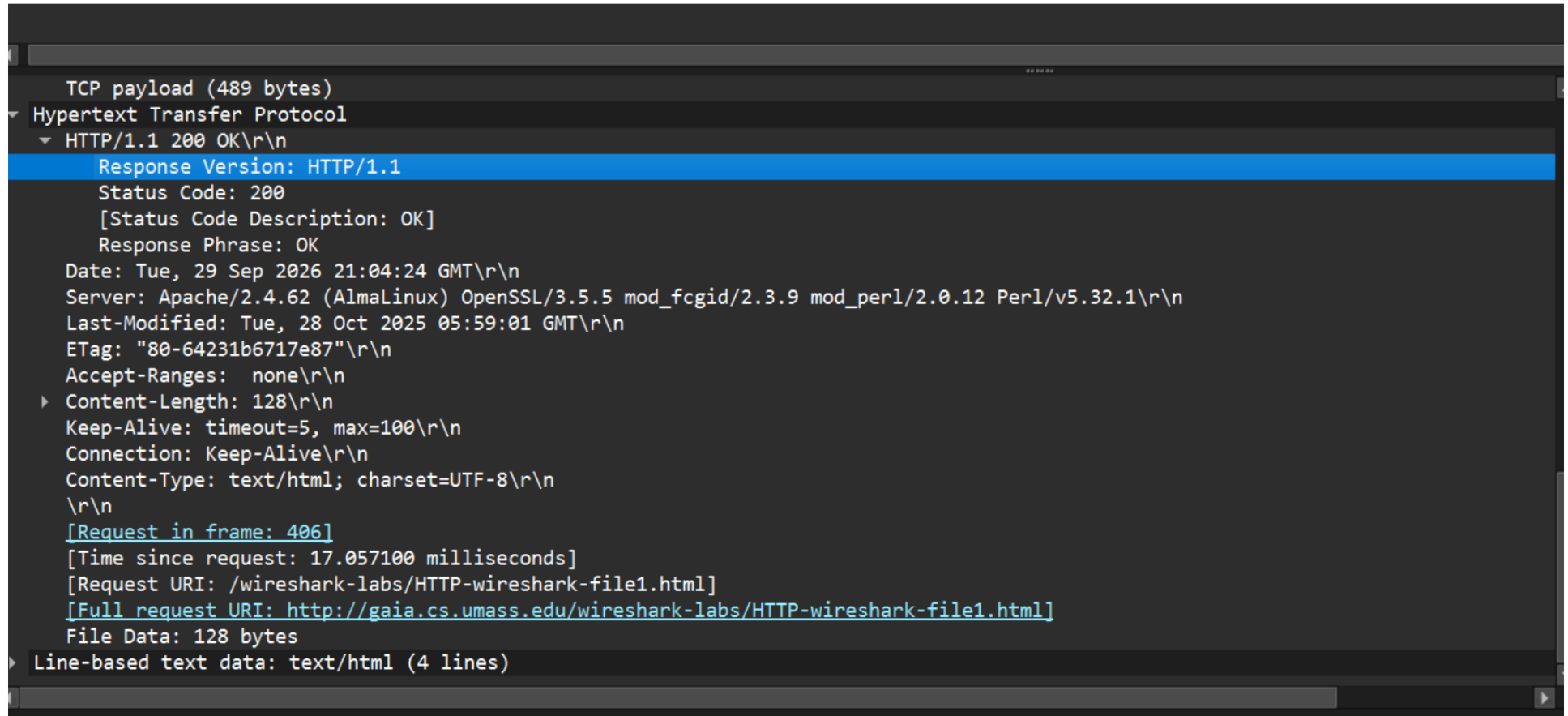
```
TCP payload (387 bytes)
Hypertext Transfer Protocol
  GET /wireshark-labs/HTTP-wireshark-file1.html HTTP/1.1\r\n
    Request Method: GET
    Request URI: /wireshark-labs/HTTP-wireshark-file1.html
    Request Version: HTTP/1.1
    Host: gaia.cs.umass.edu\r\n
  Accept: text/html,application/xhtml+xml,application/xml;q=0.9,*/*;q=0.8\r\n
  Accept-Language: en-US,en;q=0.9\r\n
  Accept-Encoding: gzip, deflate\r\n
  Connection: keep-alive\r\n
  Upgrade-Insecure-Requests: 1\r\n
  Priority: u=0, i\r\n
  \r\n
  [Response in frame: 413]
  [Full request URI: http://gaia.cs.umass.edu/wireshark-labs/HTTP-wireshark-file1.html]
```

HTTP Wireshark Lab | Question 2

What is the status code returned from the server to your browser?

The server returned status code 200 OK, indicating that the request was successful.

The server response shows Status Code: 200 and Response Phrase: OK.



The image shows a Wireshark packet capture window. The packet list on the left shows a packet of type 'Hypertext Transfer Protocol' with status 'HTTP/1.1 200 OK\r\n'. The packet details pane on the right shows the following information:

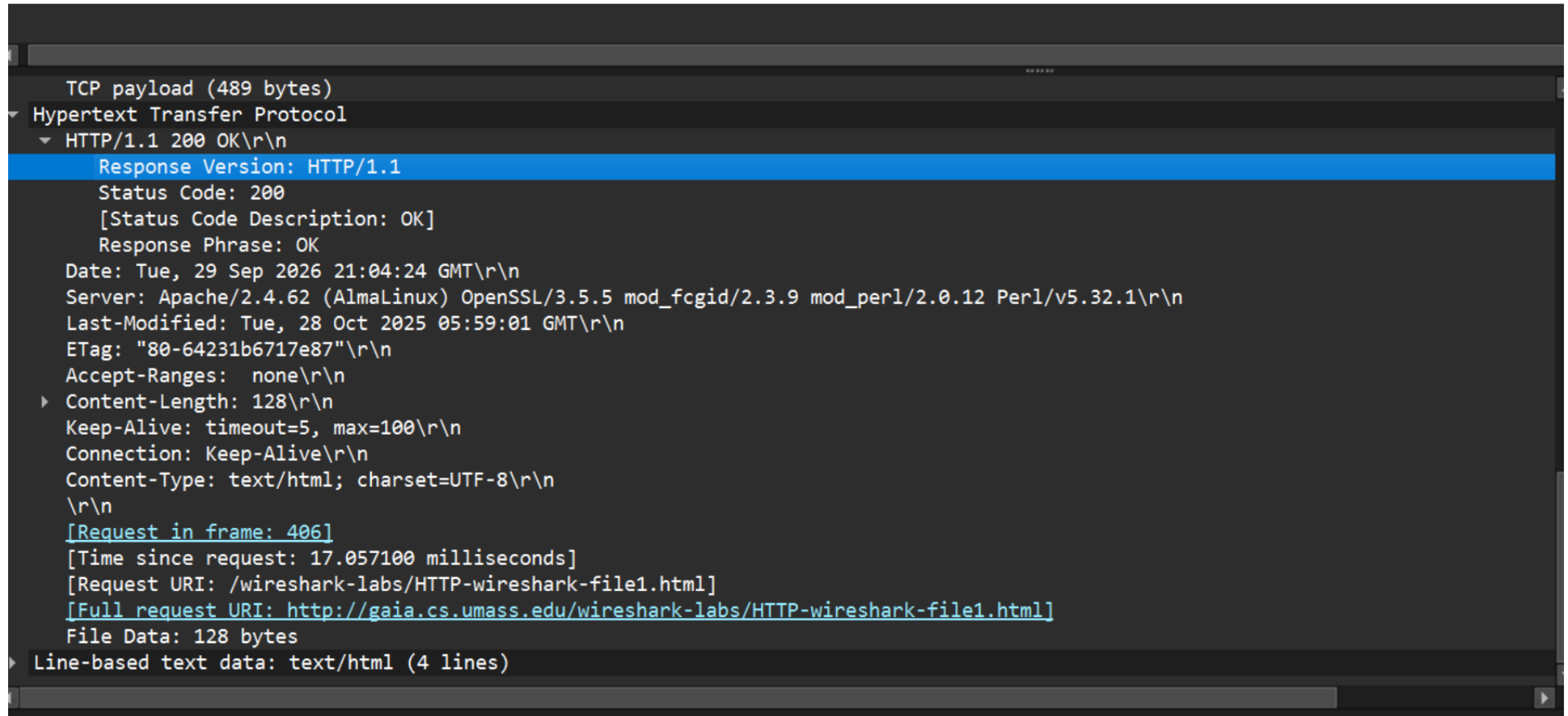
```
TCP payload (489 bytes)
Hypertext Transfer Protocol
  HTTP/1.1 200 OK\r\n
    Response Version: HTTP/1.1
    Status Code: 200
    [Status Code Description: OK]
    Response Phrase: OK
    Date: Tue, 29 Sep 2026 21:04:24 GMT\r\n
    Server: Apache/2.4.62 (AlmaLinux) OpenSSL/3.5.5 mod_fcgid/2.3.9 mod_perl/2.0.12 Perl/v5.32.1\r\n
    Last-Modified: Tue, 28 Oct 2025 05:59:01 GMT\r\n
    ETag: "80-64231b6717e87"\r\n
    Accept-Ranges: none\r\n
    Content-Length: 128\r\n
    Keep-Alive: timeout=5, max=100\r\n
    Connection: Keep-Alive\r\n
    Content-Type: text/html; charset=UTF-8\r\n
    \r\n
    [Request in frame: 406]
    [Time since request: 17.057100 milliseconds]
    [Request URI: /wireshark-labs/HTTP-wireshark-file1.html]
    [Full request URI: http://gaia.cs.umass.edu/wireshark-labs/HTTP-wireshark-file1.html]
    File Data: 128 bytes
  Line-based text data: text/html (4 lines)
```

HTTP Wireshark Lab | Question 3

When was the HTML file that you are retrieving last modified at the server?

The HTML file was last modified on Tuesday, October 28, 2025, at 05:59:01 GMT, as shown in the Last-Modified header.

Last-Modified: Tue, 28 Oct 2025 05:59:01 GMT.



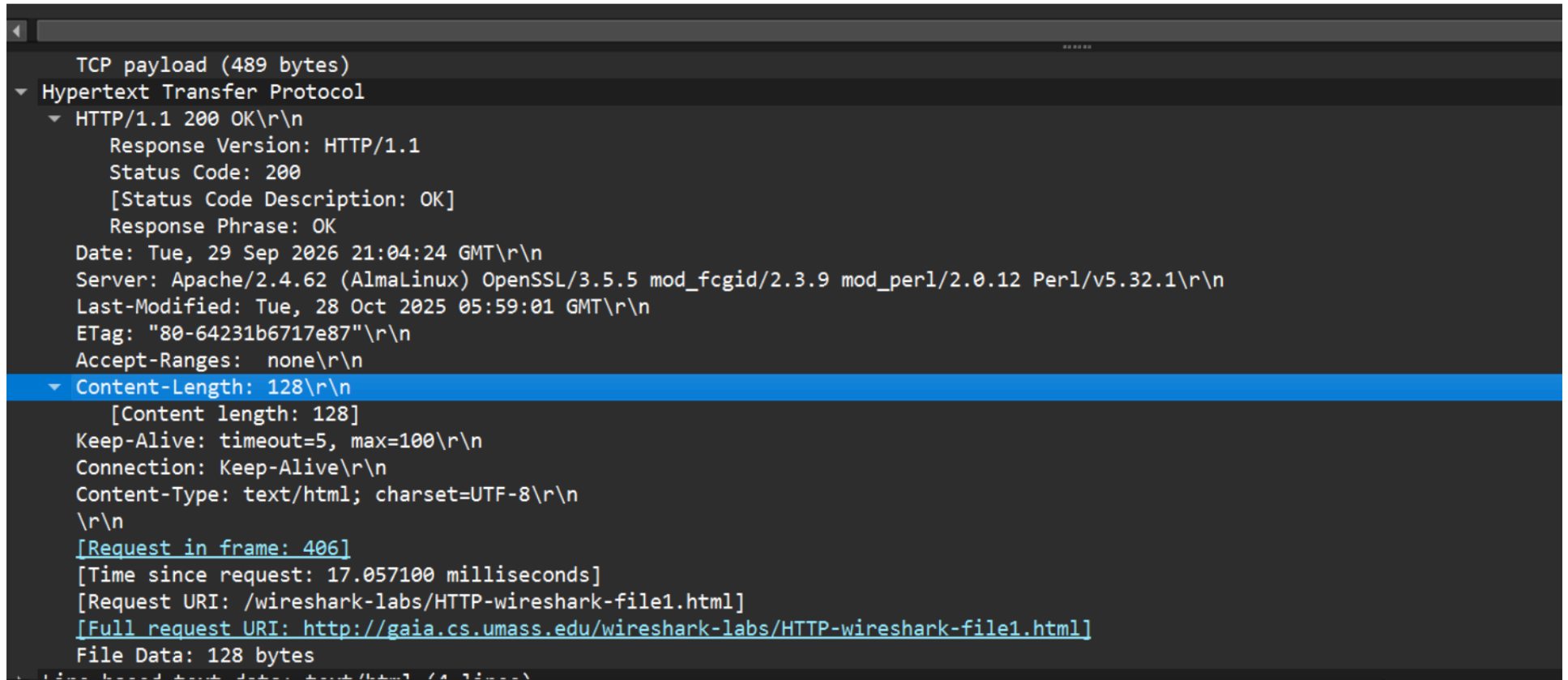
```
TCP payload (489 bytes)
Hypertext Transfer Protocol
  HTTP/1.1 200 OK\r\n
    Response Version: HTTP/1.1
    Status Code: 200
    [Status Code Description: OK]
    Response Phrase: OK
    Date: Tue, 29 Sep 2026 21:04:24 GMT\r\n
    Server: Apache/2.4.62 (AlmaLinux) OpenSSL/3.5.5 mod_fcgid/2.3.9 mod_perl/2.0.12 Perl/v5.32.1\r\n
    Last-Modified: Tue, 28 Oct 2025 05:59:01 GMT\r\n
    ETag: "80-64231b6717e87"\r\n
    Accept-Ranges: none\r\n
    Content-Length: 128\r\n
    Keep-Alive: timeout=5, max=100\r\n
    Connection: Keep-Alive\r\n
    Content-Type: text/html; charset=UTF-8\r\n
    \r\n
    [Request in frame: 406]
    [Time since request: 17.057100 milliseconds]
    [Request URI: /wireshark-labs/HTTP-wireshark-file1.html]
    [Full request URI: http://gaia.cs.umass.edu/wireshark-labs/HTTP-wireshark-file1.html]
    File Data: 128 bytes
  Line-based text data: text/html (4 lines)
```

HTTP Wireshark Lab | Question 4

How many bytes of content are being returned to your browser?

128 bytes of content were returned to my browser, as shown by Content-Length: 128.

Content-Length: 128 is highlighted in the server response.

A screenshot of the Wireshark network protocol analyzer interface. The packet list on the left shows a selected packet of type 'Hypertext Transfer Protocol'. The packet details pane on the right displays the structure of the HTTP response. The 'Content-Length: 128\r\n' field is highlighted with a blue background. Other visible fields include 'HTTP/1.1 200 OK\r\n', 'Response Version: HTTP/1.1', 'Status Code: 200', 'Date: Tue, 29 Sep 2026 21:04:24 GMT\r\n', 'Server: Apache/2.4.62 (AlmaLinux) OpenSSL/3.5.5 mod_fcgid/2.3.9 mod_perl/2.0.12 Perl/v5.32.1\r\n', 'Last-Modified: Tue, 28 Oct 2025 05:59:01 GMT\r\n', 'ETag: "80-64231b6717e87"\r\n', 'Accept-Ranges: none\r\n', 'Keep-Alive: timeout=5, max=100\r\n', 'Connection: Keep-Alive\r\n', 'Content-Type: text/html; charset=UTF-8\r\n', and 'File Data: 128 bytes'.

```
TCP payload (489 bytes)
Hypertext Transfer Protocol
  HTTP/1.1 200 OK\r\n
    Response Version: HTTP/1.1
    Status Code: 200
    [Status Code Description: OK]
    Response Phrase: OK
    Date: Tue, 29 Sep 2026 21:04:24 GMT\r\n
    Server: Apache/2.4.62 (AlmaLinux) OpenSSL/3.5.5 mod_fcgid/2.3.9 mod_perl/2.0.12 Perl/v5.32.1\r\n
    Last-Modified: Tue, 28 Oct 2025 05:59:01 GMT\r\n
    ETag: "80-64231b6717e87"\r\n
    Accept-Ranges: none\r\n
  Content-Length: 128\r\n
    [Content length: 128]
    Keep-Alive: timeout=5, max=100\r\n
    Connection: Keep-Alive\r\n
    Content-Type: text/html; charset=UTF-8\r\n
    \r\n
    [Request in frame: 406]
    [Time since request: 17.057100 milliseconds]
    [Request URI: /wireshark-labs/HTTP-wireshark-file1.html]
    [Full request URI: http://gaia.cs.umass.edu/wireshark-labs/HTTP-wireshark-file1.html]
    File Data: 128 bytes
```

HTTP Wireshark Lab | Question 5

Inspect the contents of the first HTTP GET request from your browser to the server. Do you see an "IF-MODIFIED-SINCE" line in the HTTP GET?

No. The first HTTP GET request, in packet 957, does not contain an If-Modified-Since header.

Packet 957 shows the complete first GET request headers.

No.	Time	Source	Destination	Protocol	Length	Info
957	6.827191400			HTTP	441	GET /wireshark-labs/HTTP-wireshark-file2.html HTTP/1.1
959	6.848149600			HTTP	787	HTTP/1.1 200 OK (text/html)
962	6.957078400			HTTP	461	GET /favicon.ico HTTP/1.1
967	6.976111500			HTTP	636	HTTP/1.1 301 Moved Permanently (text/html)
1053	10.360537200			HTTP	527	GET /wireshark-labs/HTTP-wireshark-file2.html HTTP/1.1
1055	10.581953500			HTTP	364	HTTP/1.1 304 Not Modified

Transmission Control Protocol, Src Port: 59789, Dst Port: 80, Seq: 1, Ack: 1, Len: 387

Hypertext Transfer Protocol

- GET /wireshark-labs/HTTP-wireshark-file2.html HTTP/1.1\r\n
- Host: gaia.cs.umass.edu\r\n
- Accept: text/html,application/xhtml+xml,application/xml;q=0.9,*/*;q=0.8\r\n
- Accept-Language: en-US,en;q=0.9\r\n
- Accept-Encoding: gzip, deflate\r\n
- Connection: keep-alive\r\n
- Upgrade-Insecure-Requests: 1\r\n
- Priority: u=0, i\r\n
- \r\n

[Response in frame: 959]

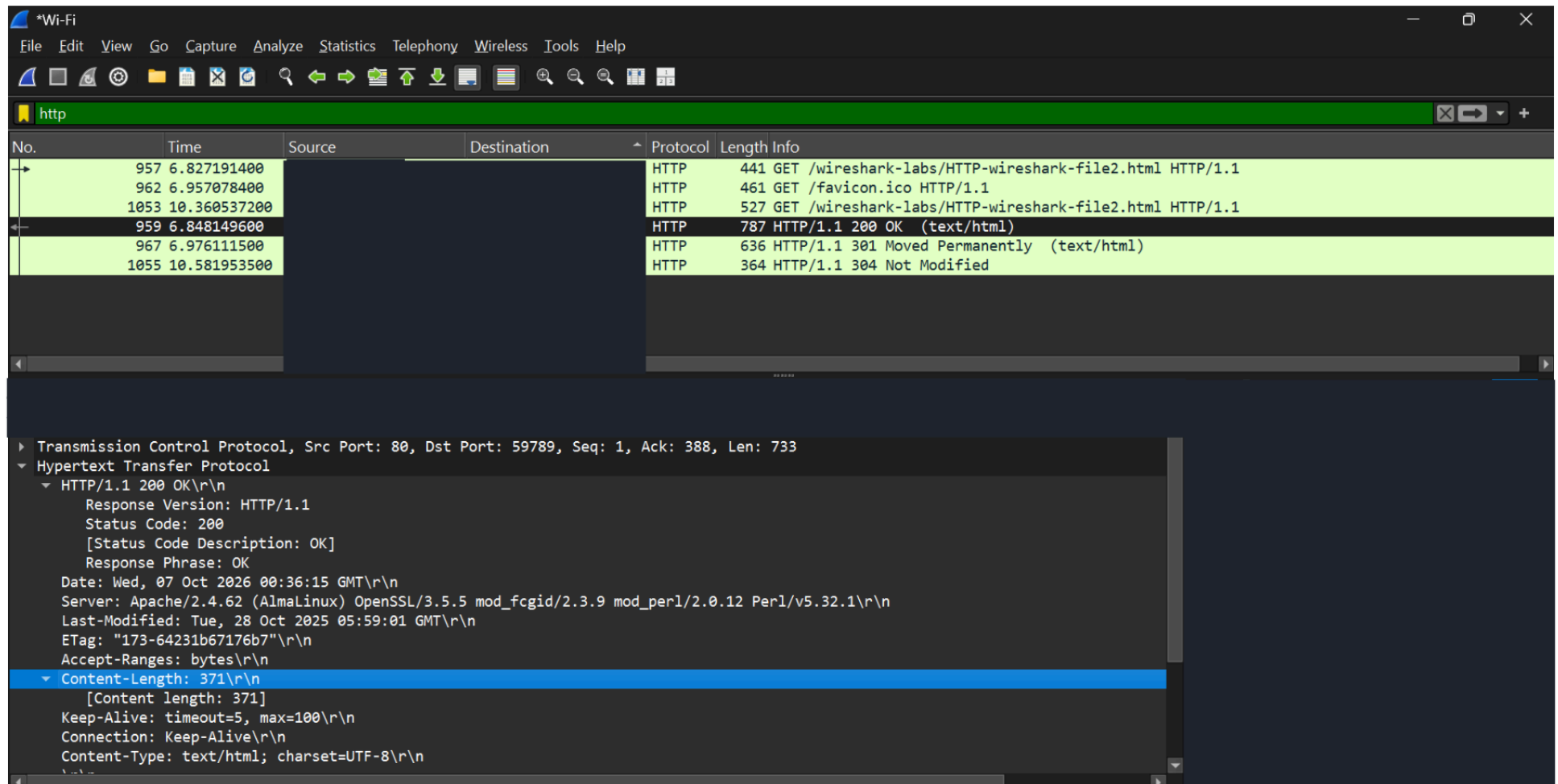
[Full request URI: <http://gaia.cs.umass.edu/wireshark-labs/HTTP-wireshark-file2.html>]

HTTP Wireshark Lab | Question 6

Inspect the contents of the server response. Did the server explicitly return the contents of the file? How can you tell?

Yes. The first server response, in packet 959, returned the HTML file contents. It shows HTTP/1.1 200 OK and Content-Length: 371, indicating that 371 bytes of HTML content were returned.

Packet 959: the first response is 200 OK, with Content-Length: 371 highlighted.



The image shows a Wireshark packet capture window. The top pane displays a list of packets. Packet 959 is selected, showing an HTTP 200 OK response. The bottom pane shows the details of packet 959, including the Hypertext Transfer Protocol section. The 'Content-Length: 371' field is highlighted in blue.

No.	Time	Source	Destination	Protocol	Length	Info
957	6.827191400			HTTP	441	GET /wireshark-labs/HTTP-wireshark-file2.html HTTP/1.1
962	6.957078400			HTTP	461	GET /favicon.ico HTTP/1.1
1053	10.360537200			HTTP	527	GET /wireshark-labs/HTTP-wireshark-file2.html HTTP/1.1
959	6.848149600			HTTP	787	HTTP/1.1 200 OK (text/html)
967	6.976111500			HTTP	636	HTTP/1.1 301 Moved Permanently (text/html)
1055	10.581953500			HTTP	364	HTTP/1.1 304 Not Modified

Transmission Control Protocol, Src Port: 80, Dst Port: 59789, Seq: 1, Ack: 388, Len: 733

Hypertext Transfer Protocol

- HTTP/1.1 200 OK\r\n
 - Response Version: HTTP/1.1
 - Status Code: 200
 - [Status Code Description: OK]
 - Response Phrase: OK
 - Date: Wed, 07 Oct 2026 00:36:15 GMT\r\n
 - Server: Apache/2.4.62 (AlmaLinux) OpenSSL/3.5.5 mod_fcgid/2.3.9 mod_perl/2.0.12 Perl/v5.32.1\r\n
 - Last-Modified: Tue, 28 Oct 2025 05:59:01 GMT\r\n
 - ETag: "173-64231b67176b7"\r\n
 - Accept-Ranges: bytes\r\n
 - Content-Length: 371\r\n**
 - [Content length: 371]
 - Keep-Alive: timeout=5, max=100\r\n
 - Connection: Keep-Alive\r\n
 - Content-Type: text/html; charset=UTF-8\r\n

HTTP Wireshark Lab | Question 7

Now inspect the contents of the second HTTP GET request from your browser to the server. Do you see an "IF-MODIFIED-SINCE:" line in the HTTP GET? If so, what information follows the "IF-MODIFIED-SINCE:" header?

Yes. The second GET request, in packet 1053, contains If-Modified-Since: Tue, 28 Oct 2025 05:59:01 GMT. This is the last-modified date associated with the browser's cached copy of the file.

Packet 1053: the If-Modified-Since header is highlighted.

The screenshot shows the Wireshark network protocol analyzer interface. The top pane displays a list of captured packets. Packet 1053, an HTTP GET request, is selected and highlighted. The middle pane shows the details of the selected packet, with the 'Hypertext Transfer Protocol' section expanded. Within this section, the 'If-Modified-Since' header is highlighted in blue. The bottom pane shows the raw packet data in hexadecimal and ASCII.

No.	Time	Source	Destination	Protocol	Length	Info
957	6.827191400			HTTP	441	GET /wireshark-labs/HTTP-wireshark-file2.html HTTP/1.1
962	6.957078400			HTTP	461	GET /favicon.ico HTTP/1.1
1053	10.360537200			HTTP	527	GET /wireshark-labs/HTTP-wireshark-file2.html HTTP/1.1
959	6.848149600			HTTP	787	HTTP/1.1 200 OK (text/html)
967	6.976111500			HTTP	636	HTTP/1.1 301 Moved Permanently (text/html)
1055	10.581953500			HTTP	364	HTTP/1.1 304 Not Modified

Transmission Control Protocol, Src Port: 59789, Dst Port: 80, Seq: 795, Ack: 1316, Len: 473

Hypertext Transfer Protocol

- GET /wireshark-labs/HTTP-wireshark-file2.html HTTP/1.1\r\n
- Request Method: GET
- Request URI: /wireshark-labs/HTTP-wireshark-file2.html
- Request Version: HTTP/1.1
- Host: gaia.cs.umass.edu\r\n
- Accept: text/html,application/xhtml+xml,application/xml;q=0.9,*/*;q=0.8\r\n
- Accept-Language: en-US,en;q=0.9\r\n
- Accept-Encoding: gzip, deflate\r\n
- Connection: keep-alive\r\n
- Upgrade-Insecure-Requests: 1\r\n
- If-Modified-Since: Tue, 28 Oct 2025 05:59:01 GMT\r\n**
- If-None-Match: "173-64231b67176b7"\r\n
- Priority: u=0, i\r\n
- \r\n
- [Response in frame: 1055]
- [Full request URI: http://gaia.cs.umass.edu/wireshark-labs/HTTP-wireshark-file2.html]

HTTP Wireshark Lab | Question 8

What is the HTTP status code and phrase returned from the server in response to this second HTTP GET? Did the server explicitly return the contents of the file? Explain.

The server returned 304 Not Modified in packet 1055. It did not return the file contents; this response contains only headers. The file had not changed, so the browser used its cached copy.

Packet 1055: the response status is HTTP/1.1 304 Not Modified.

The image shows a Wireshark packet capture of an HTTP session. The packet list pane at the top shows several packets, with packet 1055 selected. The packet details pane below shows the structure of the selected packet, which is an HTTP response.

No.	Time	Source	Destination	Protocol	Length	Info
957	6.827191400			HTTP	441	GET /wireshark-labs/HTTP-wireshark-file2.html HTTP/1.1
962	6.957078400			HTTP	461	GET /favicon.ico HTTP/1.1
1053	10.360537200			HTTP	527	GET /wireshark-labs/HTTP-wireshark-file2.html HTTP/1.1
959	6.848149600			HTTP	787	HTTP/1.1 200 OK (text/html)
967	6.976111500			HTTP	636	HTTP/1.1 301 Moved Permanently (text/html)
1055	10.581953500			HTTP	364	HTTP/1.1 304 Not Modified

Transmission Control Protocol, Src Port: 80, Dst Port: 59789, Seq: 1316, Ack: 1268, Len: 310

Hypertext Transfer Protocol

- HTTP/1.1 304 Not Modified\r\n
 - Response Version: HTTP/1.1
 - Status Code: 304
 - [Status Code Description: Not Modified]
 - Response Phrase: Not Modified
 - Date: Wed, 07 Oct 2026 00:36:19 GMT\r\n
 - Server: Apache/2.4.62 (AlmaLinux) OpenSSL/3.5.5 mod_fcgid/2.3.9 mod_perl/2.0.12 Perl/v5.32.1\r\n
 - Last-Modified: Tue, 28 Oct 2025 05:59:01 GMT\r\n
 - ETag: "173-64231b67176b7"\r\n
 - Accept-Ranges: bytes\r\n
 - Keep-Alive: timeout=5, max=98\r\n
 - Connection: Keep-Alive\r\n
 - \r\n
 - [Request in frame: 1053]
 - [Time since request: 221.416300 milliseconds]

HTTP Wireshark Lab | Question 9

How many HTTP GET request messages did your browser send?

My browser sent 4 HTTP GET requests in the displayed capture: packet 136 for the HTML file, packet 148 for /pearson.png, packet 177 for /8E_cover_small.jpg, and packet 200 for /favicon.ico. The page and its two embedded images account for 3 requests; the favicon adds 1.

The packet list shows all four GET requests. Packet 136 also shows TCP source port 56330.

Wireshark packet capture showing HTTP GET requests. The packet list shows four GET requests: packet 136 (HTML), packet 148 (pearson.png), packet 177 (8E_cover_small.jpg), and packet 200 (favicon.ico). The packet details for packet 136 show the full request URI: http://gaia.cs.umass.edu/wireshark-labs/HTTP-wireshark-file4.html.

No.	Time	Source	Destination	Protocol	Length	Info
136	6.377210700			HTTP	441	GET /wireshark-labs/HTTP-wireshark-file4.html HTTP/1.1
139	6.921394300			HTTP	1358	HTTP/1.1 200 OK (text/html)
148	7.044743000			HTTP	464	GET /pearson.png HTTP/1.1
153	7.078120300			HTTP	636	HTTP/1.1 301 Moved Permanently (text/html)
177	7.232578400			HTTP	451	GET /8E_cover_small.jpg HTTP/1.1
200	7.360890700			HTTP	461	GET /favicon.ico HTTP/1.1
214	7.380317700			HTTP	636	HTTP/1.1 301 Moved Permanently (text/html)
282	8.353326100			HTTP	492	HTTP/1.1 200 OK (JPEG JFIF image)

Transmission Control Protocol, Src Port: 56330, Dst Port: 80, Seq: 1, Ack: 1, Len: 387

Hypertext Transfer Protocol

- GET /wireshark-labs/HTTP-wireshark-file4.html HTTP/1.1\r\n
- Request Method: GET
- Request URI: /wireshark-labs/HTTP-wireshark-file4.html
- Request Version: HTTP/1.1
- Host: gaia.cs.umass.edu\r\n
- Accept: text/html,application/xhtml+xml,application/xml;q=0.9,*/*;q=0.8\r\n
- Accept-Language: en-US,en;q=0.9\r\n
- Accept-Encoding: gzip, deflate\r\n
- Connection: keep-alive\r\n
- Upgrade-Insecure-Requests: 1\r\n
- Priority: u=0, i\r\n
- \r\n

[Response in frame: 139]
[Full request URI: http://gaia.cs.umass.edu/wireshark-labs/HTTP-wireshark-file4.html]

HTTP Wireshark Lab | Question 10

Was the connection persistent or non-persistent? How do you know?

The connection to `gaia.cs.umass.edu` was persistent. The HTML request (packet 136) and the logo request (packet 148) reused the same TCP connection, from [redacted]:56330 to 128.119.245.12:80. The requests also show Connection: keep-alive.

Packet 148 uses source port 56330 and shows keep-alive. Compare packet 136 in the Q9 screenshot.

The screenshot shows a Wireshark packet capture of an HTTP session. The top pane displays a list of packets, with packet 148 selected. The middle pane shows the packet details for the selected packet, and the bottom pane shows the raw packet data.

No.	Time	Destination	Protocol	Length	Info
136	6.377210700	[redacted]	HTTP	441	GET /wireshark-labs/HTTP-wireshark-file4.html HTTP/1.1
139	6.921394300	[redacted]	HTTP	1358	HTTP/1.1 200 OK (text/html)
148	7.044743000	[redacted]	HTTP	464	GET /pearson.png HTTP/1.1
153	7.078120300	[redacted]	HTTP	636	HTTP/1.1 301 Moved Permanently (text/html)
177	7.232578400	[redacted]	HTTP	451	GET /8E_cover_small.jpg HTTP/1.1
200	7.360890700	[redacted]	HTTP	461	GET /favicon.ico HTTP/1.1
214	7.380317700	[redacted]	HTTP	636	HTTP/1.1 301 Moved Permanently (text/html)
282	8.353326100	[redacted]	HTTP	492	HTTP/1.1 200 OK (JPEG JFIF image)

Transmission Control Protocol, Src Port: 56330, Dst Port: 80, Seq: 388, Ack: 1305, Len: 410

Hypertext Transfer Protocol

- GET /pearson.png HTTP/1.1\r\n
 - Request Method: GET
 - Request URI: /pearson.png
 - Request Version: HTTP/1.1
 - Host: gaia.cs.umass.edu\r\n
 - Accept: image/gif, image/jpeg, image/png, image/svg+xml, */*;q=0.9\r\n
 - Accept-Language: en-US,en;q=0.9\r\n
 - Accept-Encoding: gzip, deflate\r\n
 - Connection: keep-alive\r\n
 - Referer: http://gaia.cs.umass.edu/wireshark-labs/HTTP-wireshark-file4.html\r\n
 - Priority: u=4, i\r\n
 - \r\n

[Response in frame: 153]
[Full request URI: http://gaia.cs.umass.edu/pearson.png]